CHAPTER 1

INTRODUCTION AND OVERVIEW

WHAT WILL WE LEARN?

- What is image processing?
- What are the main applications of image processing?
- What is an image?
- What is a digital image?
- What are the goals of image processing algorithms?
- What are the most common image processing operations?
- Which hardware and software components are typically needed to build an image processing system?
- What is a machine vision system (MVS) and what are its main components?
- Why is it so hard to emulate the performance of the human visual system (HVS) using cameras and computers?

Human Activities Covered by Image Processing

- Medical Applications
- Industrial Applications
- Military Applications
- Law Enforcement and Security
- Consumer Electronics
- The Internet, WWW

What is an Image?

- An image is a visual usually 2-D representation of an object, a person, or a scene produced by an optical device such as a mirror, a lens, or a camera…

What Is a Digital Image?

- A digital image is a representation of a 2D image using a finite number of points, referred to as picture elements, pels, or pixels, with a finite number of colors!

What Is Digital Image Processing?

- Science of modifying digital images by means of a digital computer...
  - Images are changed automatically according to an algorithm as opposed to image manipulation using tools like Photoshop, GIMP, etc.

What Is the Scope of Image Processing?

- Low Level

  Primitive operations (e.g., noise reduction, contrast enhancement,etc.) where both the input and the output are images

- Mid Level

  Extraction of attributes (e.g., edges, contours, regions, etc.) from images

- High Level

  Analysis and interpretation of the contents of a scene

Contributing and Overlapping Areas

- Mathematics, physics, computer science, computer, optical, and electrical engineering
- Pattern recognition, machine learning, artificial intelligence, and human vision research…

Typical Image Processing Operations

- Sharpening
- Noise Removal
- Deblurring
- Edge Extraction
- Binarization
- Blurring
- Contrast Enhancement
- Object Segmentation and Labeling

Sharpening

(a) (b)

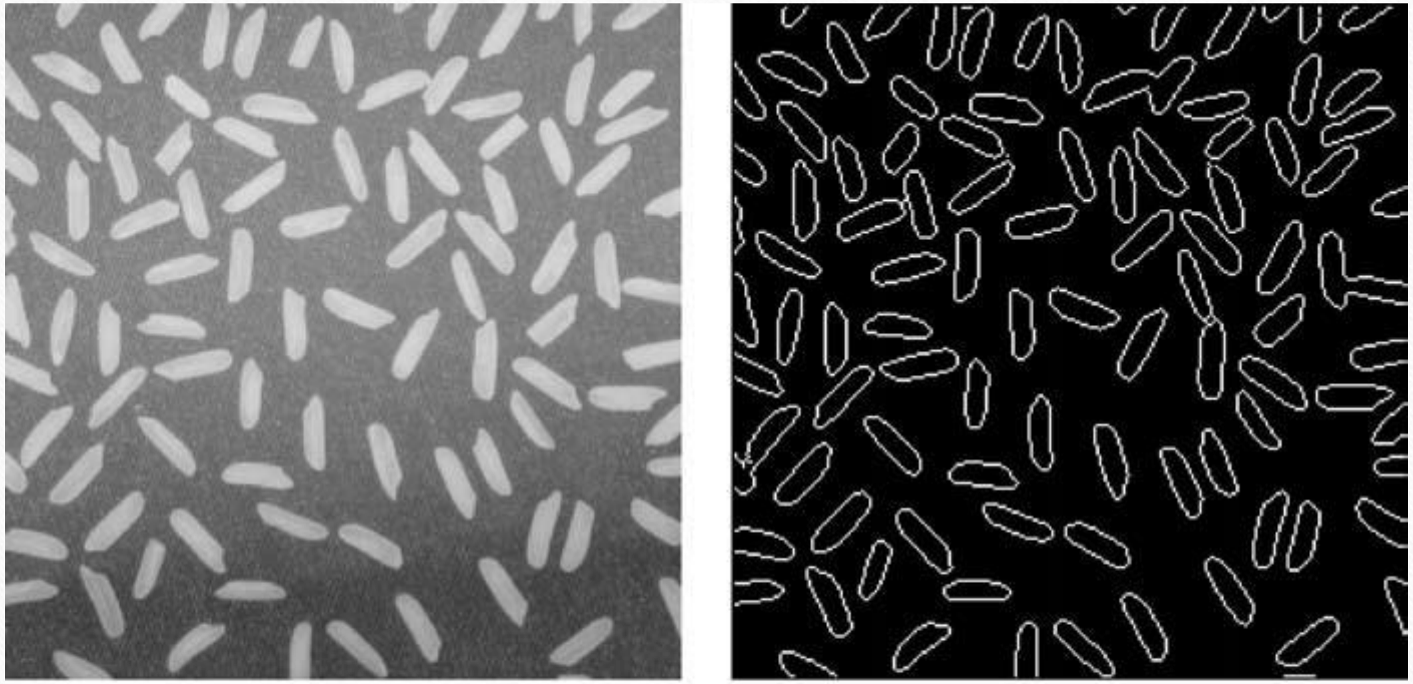
Noise Removal

(a) (b)

Deblurring

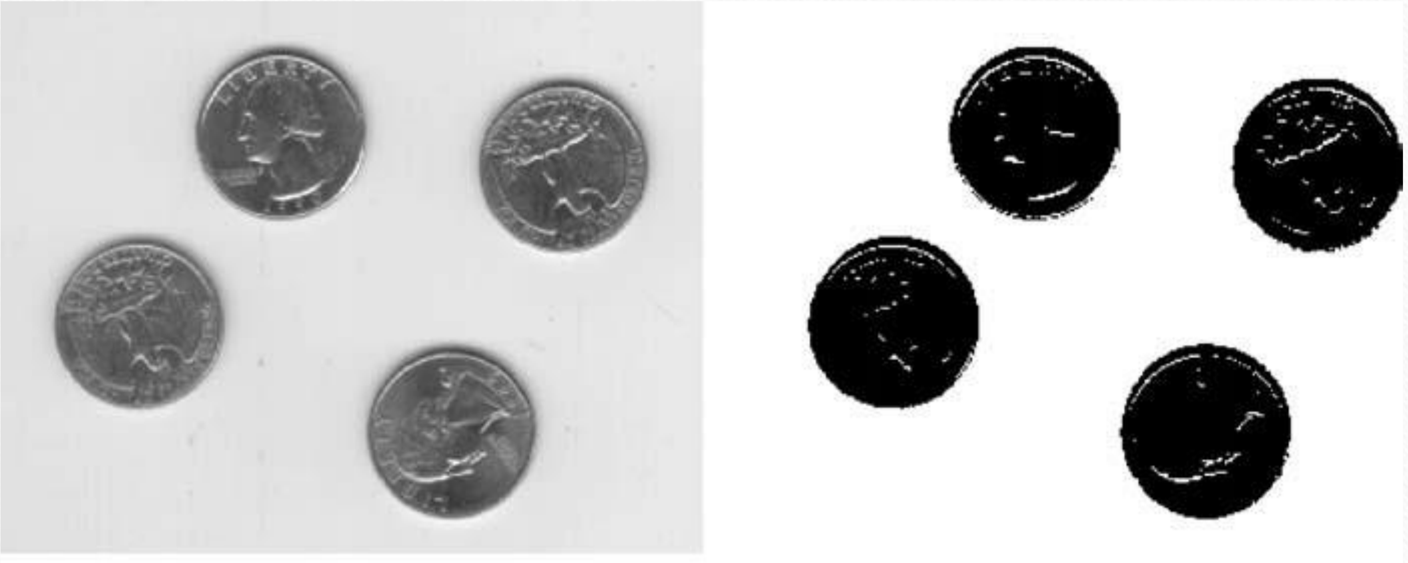
(a) (b)

Edge Extraction



(a) (b)

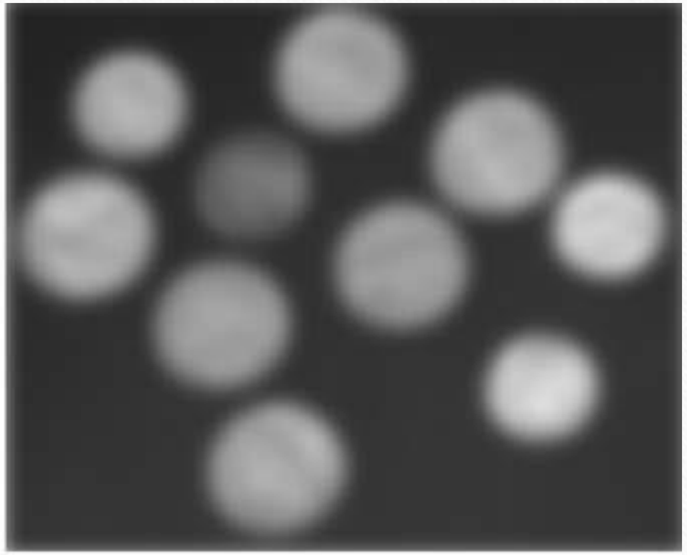
Binarization



(a) (b)

Blurring

(a)

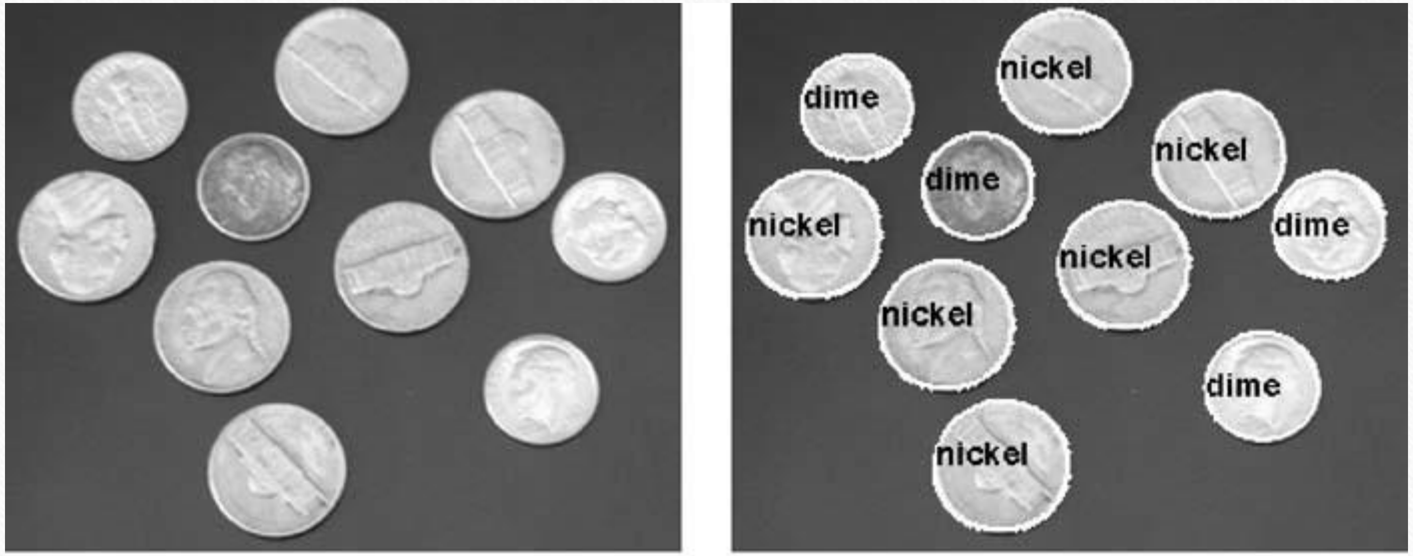


(b)

Contrast Enhancement

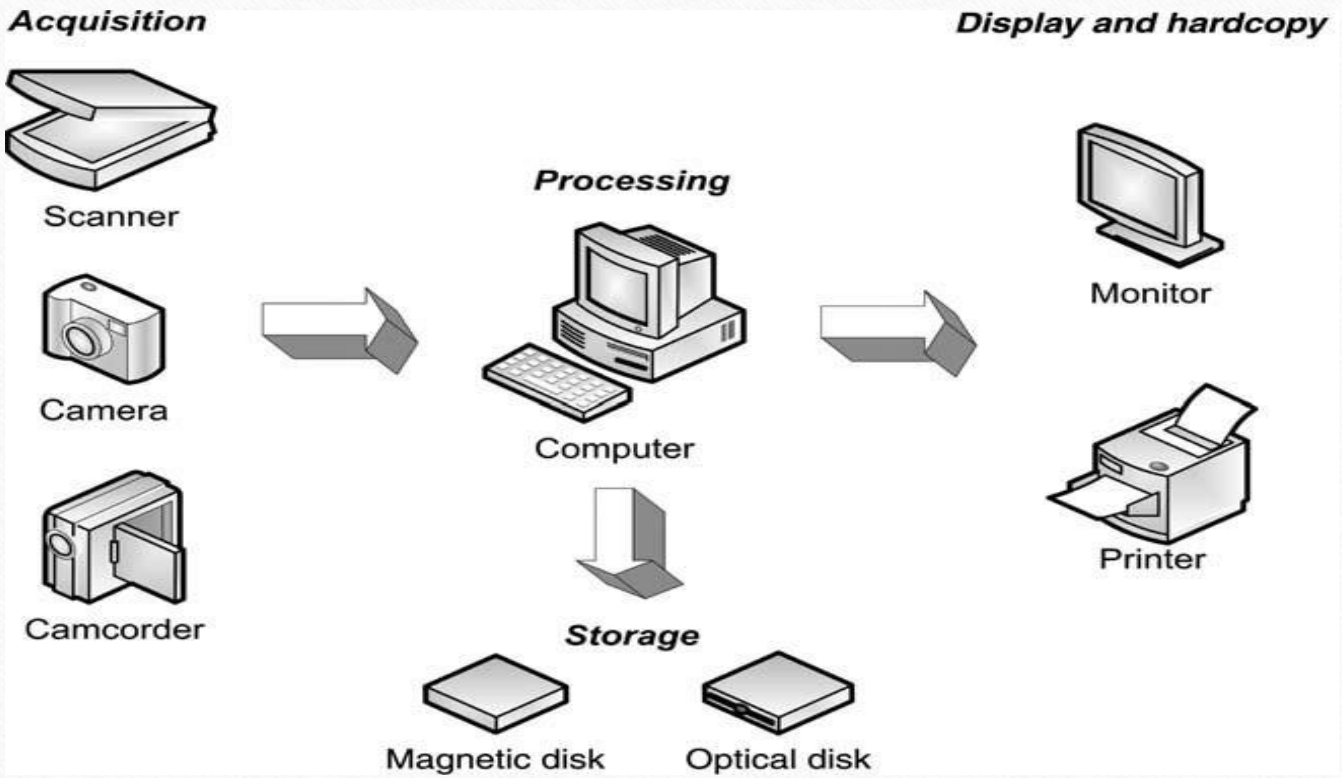
(a) (b)

Object Segmentation and Labeling



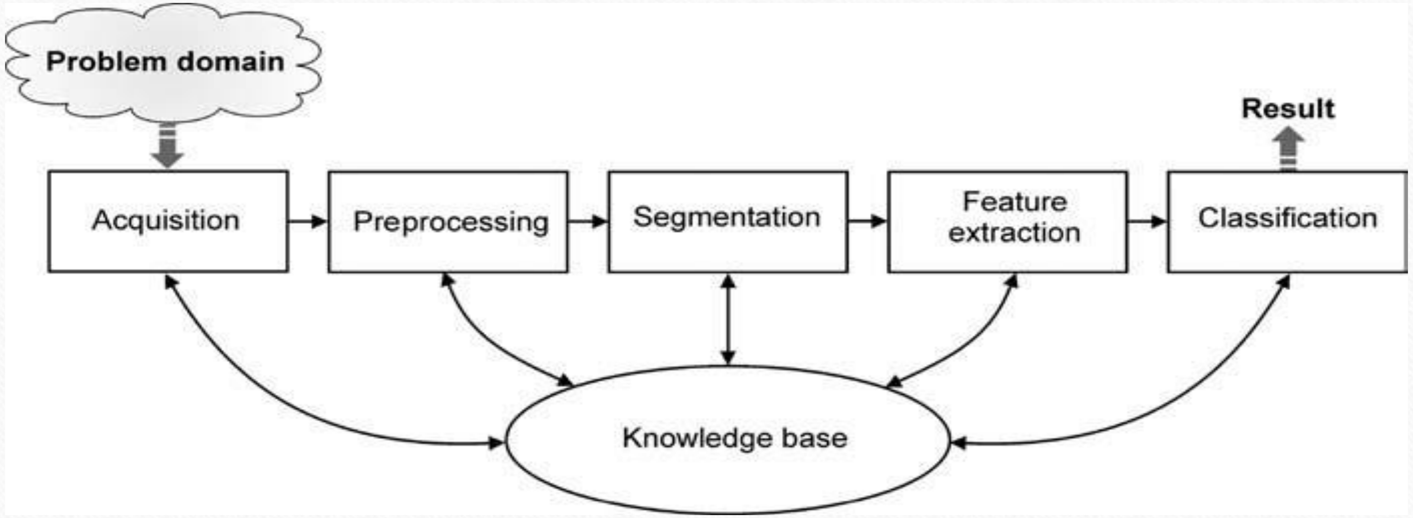
(a) (b)

Components of a Digital Image Processing System



Machine Visual System

- Sequential processing scheme of a license plate recognition system



Human Visual System

- It is extremely difficult to emulate the performance of the human visual system - in terms of processing speed, previously acquired knowledge, and the ability to resolve visual scenes under a wide range of conditions - using machine vision systems.

MVS vs. HVS

- Why is it so hard to emulate the performance of the human visual system (HVS) using cameras and computers?
  - Very large database of images and associated concepts
  - Very high speed
  - **Ability to work under a wide range of conditions**
  - Most MVS must impose numerous constraints on the operating conditions of the scene to improve their chances of success.

Easy vs Hard Problems

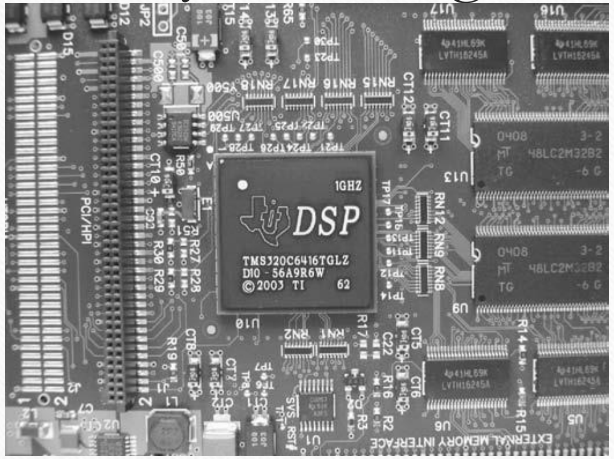
- While a computer would have a hard time identifying this as a bolt, it could easily count how many threads per inch there are :)

Easy vs Hard Problems

Easy	Hard
How wide is this plate? Is it dirty?	Look at a picture of a random kitchen, and find all the dirty plates.
Did something change between these two images?	Track an object or person moving through a crowded room of other people.
Measure the diameter of a wheel. Check to see if it is bent.	Identify arbitrary parts on pictures of bicycles.
What color is this leaf?	What kind of leaf is this?

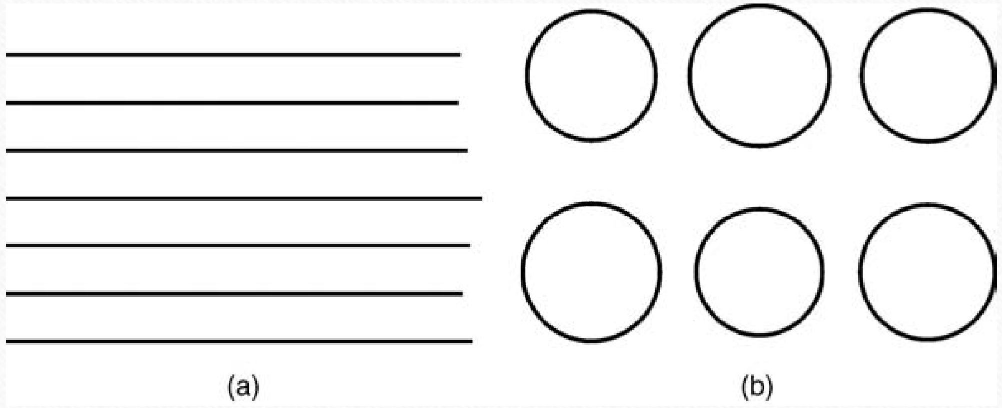
PROBLEMS

- **Prb1: Design a machine vision system** to **read the label** of the main integrated circuit (IC). Explain what each block will do, their input and output, what are the most challenging requirements, and how they will be met by the designed solution.



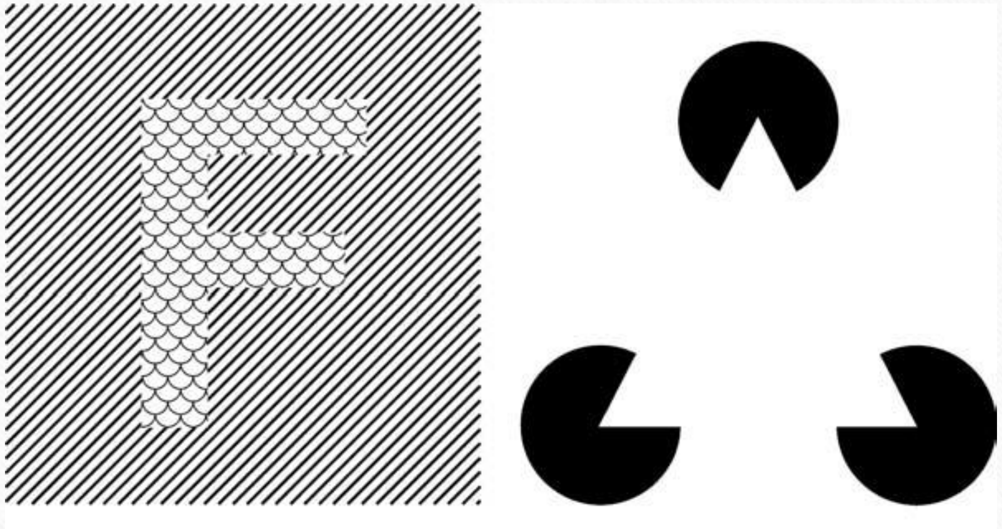
PROBLEMS

- **Prb2:** Who do you think would perform better at the following tasks- man (HVS) or computer (MVS) and why?
  a) Determining which line is the shortest
  b) Determining which circle is the smallest



PROBLEMS

- Who do you think would perform better at the following tasks:
  man (HVS) or computer (MVS) and why?
  a) Segmenting the image containing the letter “F” from the background
  b) Segmenting the white triang



(a) (b)

Topics for Assignment

References

- Practical Image and Video Processing using Matlab, Oge Marques, 2011, **Chapter 1**